SOMIL GUPTA

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SUMMARY

Computer Science undergraduate with hands-on experience in frontend engineering, graph visualization, backend systems, and transaction-safe API design using React, Next.js, TypeScript, and Node.js. Published open-source packages and contributed to production systems.

EDUCATION

Indian Institute of Information Technology Bhagalpur **Sep 2023 – Jul 2027**
*Bachelor of Technology in Computer Science and Engineering — **CGPA: 8.04/10***

EXPERIENCE

Recrivio **May 2026 – Present**
Software Development Engineer Intern *Onsite*

- Architected **Recriauth**, background verification system, applying REST API-first design and role-based access control (RBAC) via **NestJS** to handle authorization for **100%** of incoming verification requests.
- Built an in-house OCR engine using **ONNX models** and **OpenCV** for identity verification and automated PII masking, eliminating third-party API dependency and cutting operational costs by **30%**; integrated automated multi-language transcription via **FFmpeg** and scheduled cron workers.

Creova **May 2025 – Jul 2025**
Software Development Engineer Intern *Remote*

- Collaborated with cross-functional designers and engineers in Agile sprints to design and deploy **3 production-grade full-stack applications** using **Next.js, Node.js, and MongoDB**, improving scalability and reducing page load time by **35%** for **10K+ users**.
- Refined test coverage while revamping **8 Redis-cached RESTful APIs** and migrating **2.5K+ lines** of legacy JavaScript to TypeScript, cutting response latency by **40%** and lowering production bug frequency by **60%**.

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, TypeScript, SQL, Bash, Go, Rust
Frontend: React.js, Next.js, Tailwind CSS, shadcn/ui, Framer Motion
Backend: Node.js, Express.js, NestJS, FastAPI, Axum, REST APIs, WebSockets
Databases and Cloud: PostgreSQL, MongoDB, Redis, Prisma ORM, Docker, AWS (EC2, S3), CI/CD
AI and GenAI: LangChain, ONNX Models, OpenCV, OpenAI APIs, Prompt Engineering

PROJECTS

Nautilus | *Next.js, React, TypeScript, React Flow, Web Workers* **Jun 2026 – Jul 2026**

- Created a system-design visualizer with 61 architecture components, 9 templates, and interactive request-flow tracing across 13 operation types.
- Architected a Web Worker-based simulation engine with indexed graph compilation and reusable computations to model traffic, bottlenecks, queue backlogs, and infrastructure capacity.

Ledger | *TypeScript, Express 5, MongoDB, Redis, React* **May 2026 – Jun 2026**

- Modeled a double-entry ledger with atomic MongoDB transactions and conditional balance updates, preventing concurrent overdrafts and maintaining balanced journal entries.
- Implemented payment idempotency using Redis locks, response replay, and MongoDB unique indexes; Delivered authorization, capture, void, and reversal workflows with JWT rotation and RBAC.

ACHIEVEMENTS AND OPEN SOURCE

'guts-validator' (NPM Package), Jun–Jul 2026 – Authored a single-compilation schema validator in native Rust using zero-copy architecture, outperforming JS alternatives like Zod.

Open Source Contributor, Requestly (YC '24), Aug–Oct 2025 – Contributed to the core repository, enhancing API testing workflow features.

NMIT Hacks 2025 – 1st Place, May 2025 (50K+ participants) – Prototyped an AI medical diagnostic system using CNNs and multi-modal imaging in 48 hours.

Smart India Hackathon 2024 – Grand Finalist, Dec 2024 (Top 5/3K+ teams) – Devised a Deep Q-Learning traffic optimization agent with TensorFlow and SUMO, cutting wait time by **30%**.

Competitive Programming – LeetCode: 1710 rating (Top **10%**), **500+** problems solved. CodeChef 3-Star: 1710 rating.

CERTIFICATIONS

ServiceNow Certified System Administrator (CSA) **Jun 2026**
ServiceNow

Introduction to Cybersecurity **Mar 2025**
Cisco